



Gendered mode choice preferences and characteristics for educational trips in Abbottabad, Pakistan: An empirical investigation

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ABSTRACT

Educational trips are a major component of traffic in peak hours on working days. For proper trip forecasting, gendered preferences and dynamics must be understood. This study identifies gender-based model choices, preferences, and travel characteristics among school-going children and investigates the satisfaction level regarding public transport in Abbottabad, Pakistan. Using Slovin's sampling method, a total of 370 samples were collected. Descriptive statistics, chi-square tests, and paired sample t-tests were conducted to determine the significant differences between genders and their travel characteristics. For public transport satisfaction, an index was constructed using indicators extracted from a rigorous literature review. The results show a significant difference between the mode choices and preferences of boys and girls. There was no significant difference in travel time, distance, and cost, but boys tended to make more trips than girls. The satisfaction level shows that girls had a more negative opinion about public transport than boys. This study argues that transport and urban planners must incorporate both gender preferences for developing transportation alternatives for students.

1. Introduction

In Pakistan, cities are facing severe challenges in mobility and accessibility. With vehicular traffic, roads are congested, while public transit is infrequent and unsafe. The increase in travel by private vehicles and two-wheelers contributes to traffic congestion (Haider, 2014). As a result, there is a push toward encouraging public transport and reducing automobile dependency to encourage ridership (Eluru et al., 2012). Due to urbanization, population growth, and migration to urban areas, travel demand in urban regions has rapidly increased, which has led to difficulties in handling the situation effectively. Educational and work trips are the most common daily trips. Traffic congestion has occurred due to the imbalance between demand and supply, which in turn has affected urban mobility. For the facilitation of car users only, it is quite unfair to build more road infrastructure and neglect non-motorized and public transport users (Javid et al., 2014). In other developing countries, urbanization and car ownership are the main reasons for traffic congestion. Other traffic congestion factors are disorderly driving behavior, absence of traffic enforcement, inadequate infrastructure, and mismanagement of existing facilities. A rapid increase in private automobiles has made cities auto-dependent. Increased

traffic congestion affects travel delays and the driver's stress level. It also has negative environmental consequences by adding air pollution and carbon emissions. In transportation planning, the mode choice of commuters plays a vital role. Adequate mode choice analysis is needed to address forecast demand for new transportation modes, procure and allocate resources, minimize traffic congestion, and assess travel efficiencies. Many factors affect the mode choice decision, e.g., gender, age, income, car ownership, etc. (Ashalatha et al., 2013). Thus, understanding mode choice behavior is integral to reducing traffic congestion in the built environment. In peak hours, a significant percentage of trips are made for educational purposes, causing extreme traffic congestion in developing countries (Banik, Chowdhury, & Sarkar, 2009).

Educational trips have been studied in western European countries and North America, but limited studies have been conducted in the South Asian context (Singh & Vasudevan 2018). A study in Dhaka, Bangladesh, found that educational trips accounted for a large percentage of trips (Nasrin 2020). Another study investigated the mode choice of university students in both favorable and unfavorable weather conditions (Yazdani et al., 2019). The findings revealed that cost, time, comfort, and availability were the most important variables in deciding the mode of choice used for a private university student's educational

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trip. Educational trips have different characteristics compared to other trips like shopping, work-based, and recreation. Children's travel behavior varies from that of adults. The mode choice of school-going children in developing countries is still unclear. The importance of gender in travel behavior cannot be ignored, as literature shows a significant difference in the mobility patterns of both genders (Rose-nbloom, 2004). Therefore, to work out proper transportation planning, a gender perspective is imperative to understand travel behaviors, choices, and preferences.

This study aims to ascertain gender-based preferences in modal choice and satisfaction with public transport for educational-based trips. Understanding the travel patterns of students and their attitudes toward public transport services is very important for policymakers and public transport operators (Nguyen-Phuoc et al., 2018). In developing countries, such as Pakistan, understanding the travel pattern of school trips has remained unclear. Thus, the study aims to identify gender-based model choices, preferences, and travel characteristics among school-going children and investigate the satisfaction level of both genders with public transport. This study answers the following research questions (RQs):

RQ1: Do gender-based differences exist in mode choice preferences and travel characteristics?

RQ2: What are the gender-based differences with respect to the satisfaction level with public transportation?

RQ3: What is the difference between the actual and perceived satisfaction levels of students with public transport?

2. Gender-based travel behavior and public transport satisfaction

Public transportation, often known as public transportation, public transit, mass transit, or just transit, is a kind of group travel that is open to the general public, usually operated on defined routes, maintained according to a timetable, and charged a displayed fare for each journey. Due to socio-cultural and infrastructure hurdles, many women are unable to take advantage of this opportunity (Al-Rashid et al., 2020). Existing travel behavior research shows a gender gap in transportation needs and alternatives (Basarić et al., 2016). Women's travel patterns differ from men's, and these differences are marked by profound and persistent inequalities (Peters, 2013). It illustrates women's distinct and separate treatment due to the disparities in access and attitudes toward private and public transportation (Uteng, 2012). A study investigated gender experiences regarding public transportation (Al-Rashid et al., 2020). The findings found that women prefer to use cabs due to a lack of gender-responsive design and infrastructure. There is an issue of privacy and a sense of fear when traveling alone or with children. Ng & Acker (2018) studied gendered travel behavior in eight cities. The study examined mode choice, trip purpose, departure time, and travel distance in Auckland, Hanoi, Jakarta, Helsinki, Kuala Lumpur, Dublin, Manila, and Lisbon. The result showed that females commonly traveled shorter distances and preferred public transport and taxi services. Thus, it is also important to understand whether gender differences exist in educational trips for school-going children.

Limited gender-based studies exist for educational trips. Singh & Vasudevan (2018) explored students' travel decisions in Kanpur city, India using multinomial logit regression. Results showed that the lack of infrastructure support, the absence of a public transit system, and the lack of adequate school bus services resulted in school children's reliance on other motorized modes, such as family cars and paratransit. Nguyen-Phuoc et al. (2018) identified the mode choice of university students in Danang, Vietnam. Results indicated that gender, age, and household income significantly affect their mode choice behavior. Traveling time and inefficient public transport were reasons for shifting to motorcycles. Locally, Adeel et al. (2017) examined gender differences in daily trip rates in Pakistan. They used the nationally representative dataset of the 2007 Pakistan Time-Use Survey. Results show significant

differences between men's and women's travel patterns. Women were less mobile as compared to men. On average, women make 50 percent fewer trips than boys and have 46 percent shorter travel times. The result indicates that demographic indicators oppositely affect gender mobility. For example, age, marital status, and household income significantly decrease women's mobility. Adeel and Anthony (2013) investigated gender differences in travel behavior in Pakistan. Results showed significant gender differences in travel behavior regarding trip rate, mode, duration, and purpose. Females were likelier to be less mobile, with 55 percent reporting no trip in the day. In comparison to men, women made 2.8 fewer daily trips. Women were found to rely more on automobiles than men. However, no study on gender-based preferences for educational-based trips in Pakistan could be found. Therefore, it is imperative to understand gendered differences with respect to educational trips.

Public transport satisfaction is important for sustainable public transportation. Perception of the user implies the customer's fulfillment and satisfaction with the quality of service (Börjesson and Rubensson, 2019; de Oña and de Oña, 2015; van Lierop et al., 2018). A study identified that public transport users have different needs and preferences that can affect their satisfaction, e.g., accessibility, quality and reliability, ease and comfort, security, and affordability, which affect their satisfaction level (Imam, 2014). Namgung & Akar (2014) identified gendered differences in public transportation and found that gender differences existed due to their attitudes. Women had a more negative attitude towards public transport as compared to men. Ponrahono et al. (2016) studied factors influencing passenger satisfaction in Malaysia's urban and rural areas. Results showed that trip and socioeconomic characteristics influenced the satisfaction level of commuters. The results revealed significant variances with respect to gender. Women's minimum acceptable and desired service levels were greater than men's (Arabikhan et al., 2016).

The satisfaction of the public transport systems includes various vehicle and travel characteristics, environment, and experience of users. Several studies have pointed out that satisfaction with public transportation can be observed through the smoothness of the ride, fares, cleanliness of stops and vehicles, ease of boarding and disembarking, ability to find a seat, staff behavior, safety, and security (Hu et al., 2015; Weng et al., 2018), inside temperature of vehicle (Beirão and Sarsfield Cabral, 2007), punctuality (Javid et al., 2013; Seco and Gonçalves, 2007; Weng et al., 2018), seating arrangement for boys and girls, and physical condition for the vehicle (Javid et al., 2013). Beirão & Cabral (2007) evaluated comfort across different genders and age groups. They regarded comfort in terms of comfortable and clean seats, pleasant temperatures, a small number of passengers on board, the smoothness of the ride, low vibrations, and noise levels. Meanwhile, comfort at stops means having proper shelters and benches, special facilities for people with disabilities, availability of vending machines, proper lighting, an information board, trash receptacles, phone booths, etc. (Eboli and Mazzulla, 2012). Clean vehicles promote the right public image, attract passengers, and maintain ridership. Safety and security indicators in public transport are primarily safe from crime at stops and during the journey (Eboli and Mazzulla, 2012). Therefore, it is imperative to understand gender-based satisfaction regarding public transportation for sustainable transport planning.

3. Material and methods

3.1. Study area

District Abbottabad is located in the Hazara Division of Khyber Pakhtunkhwa (KPK) province, Pakistan (Fig. 1). The district has an area of about 1,967 sq. km. The city's population comprises approximately 53 percent male and 47 percent female. Abbottabad borders Mansehra District in the North, Rawalpindi District in the South, Muzaffarabad District in the East, and Haripur District in the West. It is approximately

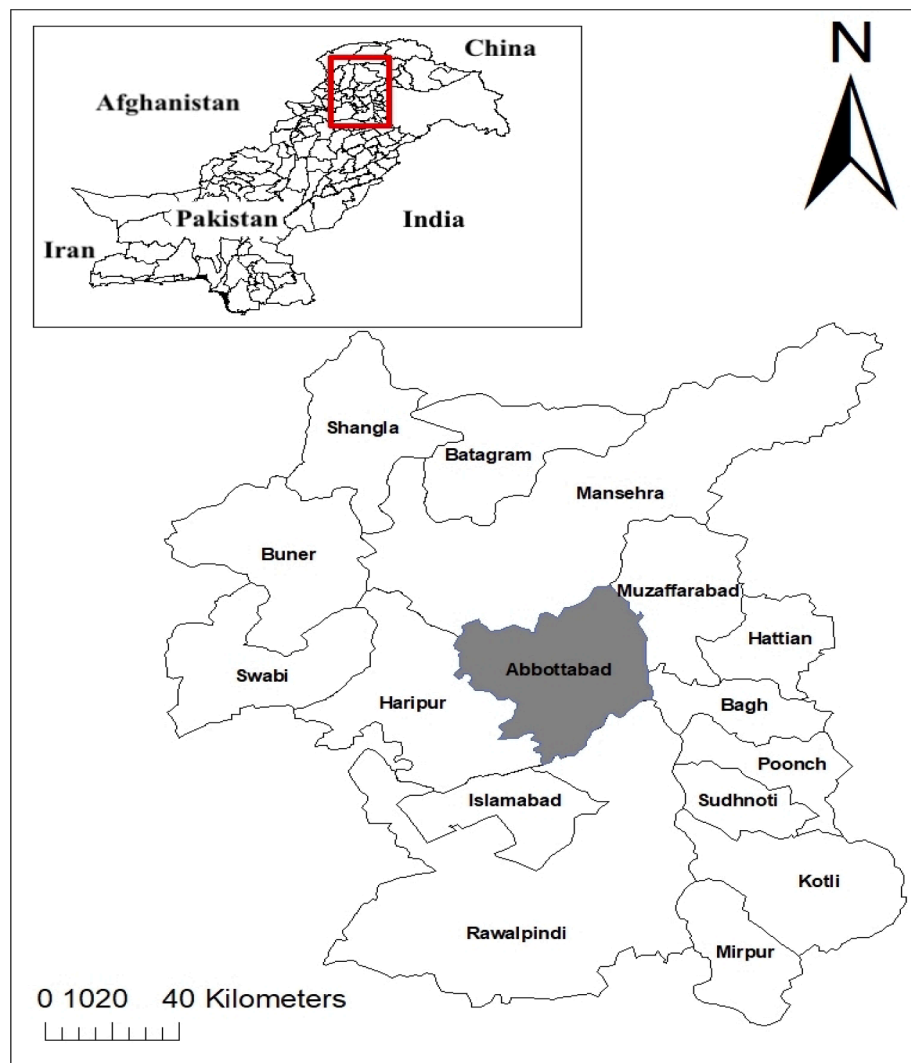


Fig. 1. Study area.

1256 m above sea level. Abbottabad faces a city with heavy rainfall all over the year. Abbottabad's average annual mean temperature is 15.9 °C, with an average of 1532 mm of rainfall each year. It is a hill station having high-standard educational institutes and is popular because of the Pakistan Military Academy, Kakul. The Abbottabad district is divided into three sub-districts (locally called tehsils), one urban municipal with 51 union councils. The city is regarded as an educational city and is locally known as the city of schools. Students from suburban areas and surrounding cities travel to the city, causing congestion during peak hours. A single arterial road serves the city, and the traffic from the surrounding cities passes through it. Therefore, it is necessary to understand the travel patterns and behavior of school-going children, which can help design transport alternatives.

In the last decade, Abbottabad city has undergone substantial land-use changes because of accelerated development, urbanization, and educational advancements (Raza et al., 2012). The city has become one of the worst cities in the country, known for frequent traffic jams that stretch for hours. After the Pakistan earthquake of 2005, Abbottabad's demography changed considerably, as a massive influx of people migrated from the surrounding area. Consequently, exponential demographic changes have increased the local population and vehicular traffic (Rashid, 2016). Rapid urbanization means demand for public and private transport has continuously increased. There is a need to meet the increase in travel demand, and it is essential to increase the use of high-

capacity vehicles and encourage citizens to use non-motorized modes. This could not be possible without knowing the travelers' behavior, their needs, and preference when using different modes. The traffic congestion problem is increasing and can only be reduced by adopting a suitable transport policy, which could not be done without improving the transport planning process in Abbottabad city.

The total number of government primary schools is 1,273 in the study areas, out of which 715 are for boys, and 558 are for girls. The cumulative number of middle schools is 166. Of those, 80 are for boys, and the rest 86 are for girls. The total number of high secondary schools is 30, 18 for boys and 12 for girls (Statistics, Bureau of Department, Planning & Development Pakhtunkhwa, 2021). There is a mixed system of public and private schools in Abbottabad. Parents decide about the school where their children should go. This decision varies according to the region and socioeconomic condition of the household. Government schools (public schools) have lower tuition fees than private schools. Both school systems have the same working hours, usually from 8 AM to 1 PM, Monday to Friday. As the timings of both public and private schools are the same, it causes congestion in peak hours.

3.2. Sampling and data collection

According to the census of 2017, the total population of Abbottabad tehsil was (N) = 981,590, and the household size was 161445. The

sample size was calculated using Slovin's Formula (Eq. 1) (Slovin 1960). With a 95 % ($e = 0.05$) confidence level, Slovin's formula gave a sample size of 400 households. Different areas were selected for the household survey, which covered almost the entire city area. The questionnaire was developed for data collection, and face-to-face interviews were conducted with children in the presence of their parents. Initially, a pilot survey was conducted to improve the questionnaire's clarity, content, and validity. A random household survey was conducted, and 22 completed questionnaires were collected. Based on respondents' feedback, some questions were modified and updated. A final standardized version of the questionnaire was prepared for a full-fledged survey. According to the sample size, 412 questionnaires were filled out by different school-going children in the presence of their family members. However, 42 questionnaires were not filled out and were discarded. As a result, the response rate was 89.8 percent. Therefore, the data from 370 questionnaires were used for further analysis. Chi-square tests, independent sample t-tests, and paired sample t-tests were conducted to find statistically significant differences between genders.

$$n = \frac{N}{1 + Ne^2} \quad (1)$$

where

- n: is the sample size.
- N: is the population size.
- e: margin of error.

3.3. Questionnaire design and selection of indicators

The following indicators were selected to measure the satisfaction level with public transport through the literature review (Table 1). Satisfaction was measured on a 1 to 5 Likert scale, where 1 is highly dissatisfied, and 5 is highly satisfied. The work was divided into three phases to achieve the goals of this study. Slovin's sampling method was used to ascertain minimum samples. The questionnaire was divided into

Table 1
Respondents' profile.

Variables	Category	Boys		Girls	
		Frequency	Percentage	Frequency	Percentage
Age	< = 9	4	2.4	0	0
	10 – 12	38	22.5	25	12.4
	13 – 15	35	20.7	74	36.8
	16 – 18	81	47.9	91	45.3
	19+	11	6.5	11	5.5
Household income (in PKRs*)	< = 50,000	35	20.7	35	17.4
	50,001 – 100,000	72	41.6	80	39.8
	100,001 – 150,000	20	11.8	32	15.9
	150,001+	42	24.9	54	26.9
	< = 5	70	41.4	80	39.8
Household size	6 – 10	86	50.9	104	51.7
	11 – 15	7	4.1	15	7.5
	16+	6	3.6	2	1
	Middle	58	34.3	31	15.4
Education level	Secondary	35	20.7	91	45.3
	High	76	45	79	39.3
	secondary				
Family system	Joint	45	26.6	39	19.4
	Nuclear	124	73.4	162	80.6
School type	Public	75	44.4	127	63.2
	Private	94	55.6	74	36.8
Number of vehicles in the house	0	41	24.3	43	21.4
	1 – 2	116	68.6	143	71.1
	3+	12	7.1	15	7.5

Source: Field Survey, 2019.

*1 USD = 156.50 (16th Oct 2019).

four parts. 1) household characteristics (household size, income, ownership of a private vehicle, number of people earning in a household, employment of parents, number of schools-going children); 2) child characteristics such as (gender, age, type of school, disability, educational level); mode; 3) travel characteristics of school-going children (usual mode choice, preferred mode choice, mode choice in bad weather conditions, travel cost, travel time, travel distance, number of trips); and 4). Public transportation satisfaction questions such as the cleanliness of stops, stability of boarding and disembarking, ability to find a seat in public transport, smoothness of the ride, fares for public transport, staff behavior, safety and security on the way, cleanliness, temperature and condition of the vehicle, and seating arrangement for boys and girls (see [supplementary materials](#)). Another indicator for overall perceived satisfaction was also asked.

The respondents were asked to choose from a set of alternatives, such as walking, two-wheeler, car, minicab, van, school bus, mini pick-up truck, and public transport. Cars, minicabs, vans, and mini pick-up trucks were classified as private transports.

3.4. Data analytical methods

Descriptive analyses were performed to identify gender-based differences in travel characteristics.

Statistical tests, like chi-square and t-tests, were performed to ascertain differences between genders. The chi-square test is used to see if a distributional outcome matches an expected pattern (Eq. 1).

$$\chi^2 = \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \quad (1)$$

An index was constructed to assess the overall satisfaction of students regarding public transport use. In order to get the actual satisfaction level, the weighted average index method was used to construct the index (Eq 2.). To get perceived satisfaction, a simple question was asked from all the respondents regarding their overall satisfaction with public transport. Paired sample t-test was used to ascertain the difference between actual and perceived satisfaction. With the t-test, users can compare the means of two measurements made on the same subject, item, or related units.

$$CI = \frac{W1 + W2 + W3 + \dots Wn}{n} \quad (2)$$

where CI is the composite index, the average weighted index method was used by giving equal weights to all the indicators and n = the total number of indicators.

4. Results and discussion

In social sciences research, respondents' personnel characteristics play a significant role in expressing their responses. This study examined a set of personal characteristics, namely, gender, age, educational level, and household income, of the 370 respondents. The sample indicates that girls' respondents were more than boys, 201 (54.3 %) were girls, and 169 (45.7 %) were boys. The majority of respondents were aged 16 to 18. The majority of boys (41.6 percent) and girls (39.8 percent) stated their monthly household income was between 50,001 – 100,000 PKR. The data discerned that the respondents belonged to the middle-income group. The nuclear family system was represented by 73.4 percent of boys and 80.6 percent of girls. Most respondents (68.6 percent boys and 71.1 percent girls) owned 1 – 2 vehicles. The respondent profile is shown in Table 1.

4.1. Mode choice and preferences

The most common mode of transportation for most school-age children (29 %) was a minicab, with a domestic car coming in second (20

%). Most of the students used minicabs for their educational trips. The minicab is locally called “Suzuki Bolan” or “Carry.” It is a common wagon family specifically designed for multi-purpose use, with a passenger capacity of 5 to 7 people. Mini pick-up trucks play a vital role in public transportation inside the city. It is locally called “Ravi Suzuki” and was also commonly used as public transport, accommodating 8–10 persons. It is easily accessible as a pick-up and drop-off service, and 18 % of students use it. About 14 percent used public transportation as a means of getting to school. Only 4 percent of children walked to school. This could be due to a lack of pedestrian safety infrastructure in the city.

A chi-square test was conducted to check the difference between genders regarding usual mode choice, preferred mode choice, and mode choice in bad weather conditions. A used alpha level of 0.01 and the following hypothesis were considered: H_0 = no significant difference between genders regarding mode choice. H_1 = there is a significant difference between genders regarding mode choice. The result shows a p -value = 0.000, less than the alpha value, so the alternative hypothesis can be accepted. The significant difference between genders regarding usual mode choice is that girl students mostly used minicabs while boys used public transport.

Results for preferred mode choice ($p = 0.000$) show a significant difference between genders. Both boy and girl students preferred their family cars, but boy students also preferred motorcycles, public transport, and ride-hailing services more than girls. Mode choice in bad weather conditions shows no significant difference between genders because both genders switch towards their family cars. Mode choice and preferences are summarized in Table 2.

Table 2
Mode choice and preferences.

Variables	Frequency		Percentage		Association between gender and mode choice Chi-Square
	Boys	Girls	Boys	Girls	
Usual mode choice					
Walk	9	4	5.3	2.0	$\chi^2 = 44.484^*$
Motorcycle	12	30	7.1	1.5	
Car	33	40	19.5	19.9	
Minicab	34	72	20.1	35.8	
Van	8	4	4.7	2.0	
School bus	15	14	8.9	7.0	
Mini pick-up truck	20	50	11.8	24.9	
Public transport	38	14	22.5	7.0	
Preferred mode choice					
Walk	5	11	3.0	5.5	$\chi^2 = 33.840^*$
Motorcycle	20	0	11.8	0	
Car	85	124	50.3	61.7	
Minicab	24	33	14.2	16.4	
Van	4	1	2.4	0.5	
School bus	15	18	8.9	9.0	
Mini pick-up truck	2	4	1.2	2.0	
Public transport	5	1	3.0	0.5	
Ride-hailing services	9	9	5.3	4.5	
Mode choice in bad weather					
Walk	1	0	0.6	0	$\chi^2 = 15.548^{**}$
Motorcycle	4	1	2.4	0.5	
Car	78	97	46.2	48.3	
Minicab	35	47	20.7	23.4	
Van	3	3	1.8	1.5	
School bus	15	16	8.9	8.0	
Mini pick-up truck	7	21	4.1	10.4	
Public transport	19	8	11.2	4.0	
Ride-hailing services	7	8	4.1	4.0	

* Significant at 1%.

** Significant at 5%.

4.2. Travel characteristics

School trips for boys and girls were between 2 and 8 km. A sampled paired t -test was conducted to check for significant differences between genders regarding travel characteristics. The following hypothesis was made for total travel length: null hypothesis (H_0) = both boy and girl share all (travel time, cost, and distance) equally for school. The alternative hypothesis (H_1) = boys and girls do not share all (travel time, cost, and distance) equally for school. The p -values from the t -test were 0.260 greater than 0.01, which rejected the alternative hypothesis and accepted the null hypothesis, i.e., boys and girls traveled similar distances to school. Half of the respondents' travel costs were between PKR 1000 to 2000, whereas 21 percent paid less than 1000, and 22 percent spent between 2000 and 3000 PKRs per month. Only 8 percent of respondents said they spent more than PKR 3000 per month on educational trips. 37 percent of the students commuted for 35 to 50 min per day. Only 7 percent of students commuted for less than 20 min per day. On average, the students traveled 54 min per day. The minimum and maximum time traveled per day were 4 and 210 min, respectively. For travel time in bad weather conditions, the test result showed a p -value = 0.592, which is greater than 0.01; accepting the null hypothesis, i.e., travel time for both boys and girls was similar in bad weather conditions. The above results show that statistically, there was a significant difference between boys' and girls' usual mode choices. So, the number of trips made each day for educational purposes could vary. The hypothesis considered whether both boys and girls made a similar number of daily trips. Results show a significant difference p -value = 0.000, which is smaller than 0.01; thus, rejecting the null hypothesis and accepting alternatives, i.e., boys and girls do not make an equal number of trips per day. The mean value of the total number of trips for boys and girls was 2.51 and 2.16, respectively, which shows that boys tend to make more trips than girls. Travel characteristics are summarized in Table 3. The study finds that some travel characteristics, like travel time, varied among the genders. However, no significant difference was observed in trip length and time.

4.2.1. Usual mode choice, preferred mode choice, and mode choice in bad weather

It was observed that respondents usually preferred another mode over their current mode. A chi-square test was conducted, and the following hypothesis was considered. H_0 = usual and preferred mode choice of students are the same. Alternative hypothesis (H_1) = usual and preferred modes of students are different. The p -value from the chi-square test was 0.000 less than 0.01; so, the null hypothesis was rejected, and an alternative hypothesis was accepted, i.e., a usual and preferred mode of students are different because most of the respondents preferred their family cars but used alternative mode of travel from home to school. Both genders preferred their family cars because of the relative safety and comfort compared to other available modes of transport. Usually, ride-hailing services were not used for educational trips in the city. Another hypothesis was considered to check whether bad weather affects the usual mode choice. H_0 = usual mode choice is unaffected by bad weather, and H_1 = usual mode choice is affected by bad weather. The significant values ($p = 0.000$ less than 0.01); showed that the usual mode choice of students was affected by bad weather because the availability of public transport was less in bad weather. Secondly, the pick and drop services (e.g., minicabs, mini pick-up trucks, and vans) were hindered due to flooded roads, which resulted in delayed arrival times to their pick-up locations. Moreover, non-motorized shifted to an alternate mode of transport. Under normal weather conditions, only 20 percent of respondents traveled by private vehicle, according to the observed pattern. In comparison, this figure increased up to 47 percent in the rainy season or bad weather, which meant that almost 27 percent of the students switched to their private cars in bad weather. The test results are summarized in Table 4.

Table 3
Travel characteristics of school-going children.

Variables	Frequency		Percentage		Mean		Standard Deviation		Difference between gender and travel characteristics t-test
	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	
Total trip length in km									
< = 2.0	24	20	14.2	10.0	4.846	5.246	3.4860	3.2732	–1.129
2.1 – 5.0	82	88	48.5	43.8					
5.1 – 8.0	45	70	26.6	34.8					
8.1 – 11.0	13	15	7.7	7.5					
11.1+	5	8	3.0	4.0					
Total travel cost in rupees per month									
< = 1000	47	32	27.8	15.9	1693.85	1931.04	1445.38	1102.22	–1.748***
1001 – 2000	84	99	49.7	49.3					
2001 – 3000	24	67	14.2	28.4					
3001 – 4000	7	6	4.1	3.0					
4001+	7	7	4.1	3.5					
Total travel time in minutes									
< = 20	13	12	7.7	6.0	55.71	52.55	29.81	26.38	1.069
21 – 35	27	32	16.0	15.5					
36 – 50	55	82	32.5	40.8					
51 – 65	31	24	18.3	11.9					
66 – 80	13	27	7.7	13.4					
Total travel time in bad weather condition									
< = 30	11	12	6.5	6.0	108.40	112.19	69.96	65.06	–0.536
31 – 60	28	32	16.6	15.9					
61 – 90	50	49	29.6	24.4					
91 – 120	34	45	20.1	22.4					
121 – 150	14	17	8.3	8.5					
151 – 180	13	21	7.7	10.4					
181+	19	25	11.2	12.4					
Total number of trips per day									
2	128	187	75.7	93	2.51	2.16	0.92	0.61	4.196*
4	39	13	23.1	6					
6	2	2	1.2	1					

* Significant at 1%.

** Significant at 5%.

*** Significant at 10%.

Table 4
Usual mode choice vs preferred mode choice and mode choice in bad weather conditions.

Variables	Chi-Square
Usual mode choice vs preferred mode choice	$\chi^2 = 407.029^*$
Usual mode choice vs mode choice in bad weather conditions	$\chi^2 = 813.004^*$

* Significant at 1%.

4.2.2. Usual travel time vs travel time in bad weather conditions

Test results from Table 3 show that bad weather affects the usual mode choice of school-going children. The result shows that the p-value was 0.000 less than 0.01, showing that the usual travel time was affected by bad weather. On regular days, the average time taken was 54 min, while the average time taken in bad weather was higher, i.e., 110 min. The primary reasons for this inconvenience are low capacities and congested roads, lack of proper rainwater drainage and sewer system, and poor road network. The results are summarized in Table 5.

4.3. Satisfaction level with the public transport

The respondents were asked how long they had to wait at the bus

Table 5
Usual travel time vs travel time in bad weather conditions.

Variable (in minutes)	Mean	Standard deviation	t-test
Usual travel time	53.99	28.007	– 19.508*
Travel time in bad weather conditions	110.46	67.278	

* Significant at 1%.

stop for public transport. Results showed that 60 percent of the respondents waited less than 5 min. Whereas 34 percent of respondents reported waiting for 5 to 15 min, around 6 percent waited longer than 15 min. Overall, commuters waited an average of 8 min. Results showed that the p-value was 0.472, and waiting time at stops was similar for both boys and girls. Almost half of the respondents reached the nearest public stop from home within 5 min. Around 34 percent of respondents took time between 5 and 15 min. Only 5 percent of the respondents took more than 25 min from home to the nearest public stops. On average, respondents took 10 min to reach the nearest stops. Results showed a p-value = 0.441 > 0.01, which shows that both boys and girls took a similar time to reach the nearest public stop from home. The results are summarized in Table 6.

The result shows that p-values for satisfaction levels with safety and security, boarding and disembarking, ride quality, odor and temperature, and seating arrangement were less than 0.05 (Table 6). This implies that significant differences between genders regarding satisfaction variables existed. Many girls were dissatisfied with the odor and temperature. It has been observed that boarding and disembarking from public transport was difficult compared for boys, possibly because of inappropriate seating arrangements in public transport. In each public vehicle, only five seats were available for girl passengers. In contrast, the boy passengers could adjust and cram inside the public vehicles. The remaining variables of satisfaction like the cleanliness of stops, staff behavior, cleanliness of public vehicles, softness and cleanliness of seats, travel time punctuality, the physical condition of public vehicles, and fares of public transport had a higher p-value greater than 0.05, which shows that is no significant difference between genders regarding satisfaction level, i.e., both boy and girl respondents had the same opinion. Overall satisfaction level percentage shows that girls had a more negative opinion about public transport than boys (Table 7).

Table 6

Total waiting time and time takes to reach the nearest public stop.

Variables	Frequency		Percentage		Mean		Standard deviation		Difference between gender and travel characteristics <i>t</i> -test
	Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls	
Total waiting time at stops (in minutes)									
< = 5	100	120	59.2	59.7	7.36	7.82	6.33	5.89	−0.720
6 – 15	56	68	33.1	33.8					
16 – 25	8	9	4.7	4.5					
26+	5	4	3.0	2.0					
Time takes to reach the nearest public stop (in minutes)									
< = 5	94	88	55.6	43.8	9.41	10.08	8.42	8.19	−0.772
6 – 15	45	81	26.6	40.3					
16 – 25	24	19	14.2	9.5					
26+	6	13	3.6	6.5					

4.4. Private car vs public transport

A paired sample *t*-test was conducted to check the significant difference in total travel time and total travel cost between public transport and private car. The result showed a *p*-value of 0.000, implying a significant difference between travel time and car and public transport costs. Compared to private cars, public transport took a longer travel time significantly. The average time taken by the private car and public transport was 50 and 80 min, respectively. Public transport took longer, probably due to a lack of proper and unauthorized stops where passengers could get in and off the vehicle at random locations. Similarly, the mean value of total travel cost for a car (2768.51) was greater than that of public transport (1219.35), which shows that the distance travel cost for the car was greater than public transport. However, this cannot be generalized due to large standard deviations in travel costs. Test results are summarized in Table 8.

4.5. Actual vs perceived satisfaction

A paired sample *t*-test was conducted to identify the difference between actual and perceived satisfaction with the public transport system. Results showed that *p*-value = 0.021 < 0.05, rejecting the null hypothesis and accepting the alternative, i.e., there was a significant difference between actual and perceived satisfaction of respondents. The mean value of perceived satisfaction (2.51) was greater than actual satisfaction (2.43), which shows that people were more dissatisfied with public transport than they perceived. The test results are summarized in Table 9.

5. Conclusion

Various studies were carried out in Western countries to understand the traveling behavior of school-going students. However, in developing countries like Pakistan, the basic dynamics of educational trips (school trips) are still unclear. This paper tried to fill the gap of limited gender-based studies. The study collected data through a household survey of school-going children from Abbottabad, Pakistan. Mode choice behavior, travel characteristics, and satisfaction with public transport were explored. The first objective of the study was to find the significant difference between usual mode choice, preferred mode choice, and mode choice in bad weather between gender. A significant difference was found in usual mode choice, preferences, and mode choice in bad weather between the gender. The results found that most girls used minicabs and family cars, while boys used public transport and minicabs for educational trips. However, both boys and girls preferred to use their family cars. This study also found that the rainy season affects students' usual mode choices and travel time. In bad weather, most respondents switch to their private cars. Respondents who use public transport on normal days also switch to other modes. No significant difference between genders was found in travel characteristics (time, distance, and cost). A gender-based significant difference has been found in

satisfaction levels with public transport. The satisfaction level showed that girls have a more negative opinion about public transport than boys. A significant difference was also observed between actual and perceived satisfaction.

The findings of this study can help better understand the students' mode choice behavior, which helps planners and engineers develop infrastructure and improve policies that encourage sustainable travel. Comparative satisfaction levels with public transport show that girls have a more negative opinion about public transport than boys. The descriptive analysis also showed that few girls used public transport for educational trips. Thus, transport planners should focus on an inclusive public transportation system to encourage public transport use.

This study has its limitations which may be covered by future research. The study did not consider any other trip, for example, work-based, shopping, or recreational. The data collected during the household survey has been limited to children of middle, secondary, and high secondary school students. Primary school and university-going students were not considered. Moreover, parental choices and preferences were not included. This study only considered household characteristics, child characteristics, mode and travel characteristics, and psychological variables, including satisfaction regarding public transport. More variables can be added in future research studies. The built environment and geographic characteristics could also be incorporated for further clarity. Advanced statistical tests could also be applied (e.g., Logit and Probit models). It is proposed that land use and spatial analyses can be performed to understand the multifaceted nature of gender-based travel behavior.

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Availability of data and material

Will be made available upon request.

CRediT authorship contribution statement

Rida Hameed Lodhi: Conceptualization, Data curation, Methodology, Formal analysis, Visualization, Writing – review & editing. **Irfan Ahmad Rana:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Abdul Waheed:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 7
Satisfaction level with public transport.

Variables	Gender	Frequency (percentage)					Difference between gender and satisfaction level Chi-Square
		Very Dissatisfied	Dissatisfied	Neutral	Satisfied	Very Satisfied	
Satisfaction level with the cleanliness of stops	Boys	39 (23.1 %)	68 (40.2 %)	33 (19.5 %)	26 (15.4 %)	3 (1.8 %)	4.194
	Girls	62 (30.8 %)	74 (36.8 %)	33 (16.4 %)	31 (15.4 %)	1 (0.5 %)	
Satisfaction level with staff behavior	Boys	22 (13.0 %)	53 (31.4 %)	45 (26.6 %)	40 (23.7 %)	9 (5.3 %)	7.111
	Girls	40 (19.9 %)	58 (28.9 %)	50 (24.9 %)	50 (24.9 %)	3 (1.5 %)	
Satisfaction level with safety and security	Boys	23 (13.6 %)	47 (27.8 %)	37 (21.9 %)	59 (34.9 %)	3 (1.8 %)	25.069*
	Girls	61 (30.3 %)	68 (33.8 %)	35 (17.4 %)	36 (17.9 %)	1 (17.9 %)	
Satisfaction level with boarding and disembarking	Boys	29 (17.2 %)	40 (23.7 %)	57 (33.7 %)	41 (24.3 %)	2 (1.2 %)	18.627*
	Girls	65 (32.3 %)	58 (28.9 %)	49 (24.4 %)	26 (12.9 %)	3 (1.5 %)	
Satisfaction level with finding seats	Boys	10 (5.9 %)	61 (36.1 %)	22 (13.0 %)	70 (41.4 %)	6 (3.6 %)	25.982*
	Girls	26 (12.9 %)	42 (20.9 %)	58 (28.9 %)	64 (31.8 %)	11 (5.5 %)	
Satisfaction level with the cleanliness of public vehicles	Boys	24 (14.2 %)	64 (37.9 %)	36 (21.3 %)	42 (24.9 %)	3 (1.8 %)	8.154***
	Girls	52 (25.9 %)	65 (32.3 %)	41 (20.4 %)	41 (20.4 %)	2 (1.0 %)	
Satisfaction level with ride quality	Boys	18 (10.7 %)	74 (43.8 %)	46 (27.2 %)	29 (17.2 %)	2 (1.2 %)	13.275*
	Girls	45 (22.4 %)	87 (43.3 %)	48 (23.9 %)	21 (10.4 %)	0 (0.0 %)	
Satisfaction level with odor and temperature	Boys	40 (23.7 %)	84 (49.7 %)	27 (16.0 %)	17 (10.1 %)	1 (0.6 %)	13.617*
	Girls	79 (39.3 %)	71 (35.3 %)	36 (17.9 %)	15 (7.5 %)	0 (0.0 %)	
Satisfaction level with softness and cleanliness of seats	Boys	40 (23.7 %)	69 (40.8 %)	34 (20.1 %)	25 (14.8 %)	1 (0.6 %)	8.674***
	Girls	57 (28.4 %)	82 (40.8 %)	49 (24.4 %)	12 (6.0 %)	1 (0.5 %)	
Satisfaction level with Punctuality of travel time	Boys	27 (16.0 %)	64 (37.9 %)	41 (24.3 %)	36 (21.3 %)	1 (0.6 %)	4.440
	Girls	48 (23.9 %)	74 (36.8 %)	44 (21.9 %)	33 (16.4 %)	2 (1.0 %)	
Satisfaction level with the physical condition of public vehicle	Boys	29 (17.2 %)	68 (40.2 %)	42 (24.9 %)	27 (16.0 %)	3 (1.8 %)	8.164***
	Girls	48 (23.9 %)	87 (43.3 %)	45 (22.4 %)	21 (10.0 %)	0 (0.0 %)	
Satisfaction level with seating arrangement for both boy and girl	Boys	45 (26.6 %)	55 (32.5 %)	22 (13.0 %)	39 (23.1 %)	8 (4.7 %)	10.101**
	Girls	80 (39.8 %)	56 (27.9 %)	29 (14.4 %)	32 (15.9 %)	4 (2.0 %)	
Satisfaction level with fares of public transport	Boys	23 (13.6 %)	49 (29.0 %)	53 (31.4 %)	38 (22.5 %)	6 (3.6 %)	0.671
	Girls	26 (12.9 %)	55 (27.4 %)	68 (33.8 %)	47 (23.4 %)	5 (2.5 %)	

* Significant at 1%.

** Significant at 5%.

*** Significant at 10%.

Table 8
Private car vs public transport.

Variable	Mean	Standard Deviation	t-test
Total travel time by car (in minutes)	50.00	24.728	−23.828*
Total travel time by public transport (in minutes)	79.97	39.462	
Total travel cost when using a car (in PKR)	2768.51	1852.023	18.158*
Total travel cost when using public transport (in PKR)	1219.35	730.213	

* Significant at 1%.

Table 9
Actual vs perceived satisfaction with public transport.

Variables	Mean	Standard Deviation	t-test
Overall actual satisfaction level	2.43	0.661	− 2.327**
Overall perceived satisfaction level	2.51	0.926	

** Significant at 5%.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cstp.2022.09.010>.

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